

Multimodal Feedback Pen Shaped Interface and MR Application with Spatial Reality Display

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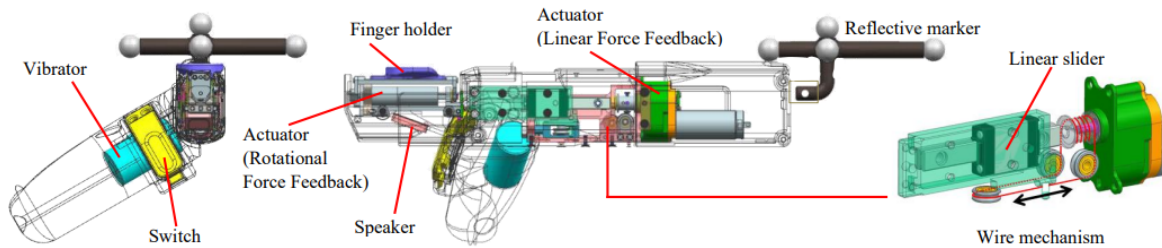


Figure 1: Mechanical structure of pen shaped interface

ABSTRACT

Multimodal interface is essential to enrich the reality of drawing in the virtual 3D environment. We propose the pen shaped interface capable of providing the following multimodal feedbacks: (1) Linear motion force feedback to express contact pressure of virtual object, (2) Rotational force feedback to simulate friction of rubbing virtual surface and tangential contact force with virtual object, (3) Vibrotactile feedback, and (4) auditory feedback to express contact information and texture of virtual object. We developed Mixed Reality (MR) interaction system with pen shaped interface and Spatial Reality Display. This system will display virtual pen tip extended from the actual pen shaped interface, and user can use it to draw into virtual workspace as well as interact with virtual objects. This demonstrates the advantage of our proposal by improving the reality of virtual space interaction.

CCS CONCEPTS

• Human-centered computing; • Human computer interaction (HCI); • Interaction devices; • Haptic devices;

KEYWORDS

Virtual Reality, Mixed Reality, Multimodal, Haptic Interface, 3D Drawing, Force Feedback, Vibrotactile, pen

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1 INTRODUCTION

The field of digital 3D drawing has been remarkably advancing with development of displays such as 3D monitor and high-quality VR headsets, as well as research of pen shaped interfaces. Drawing in the virtual space is often difficult for untrained users because of lack of physical contact information with drawing surface. There are several studies of pen type interface with haptic feedback to solve this problem. VRSketchPen provides vibrotactile and pneumatic force feedback to express contact with virtual canvas, that leads to improvement of drawing accuracy [Elsayed et al., 2020]. MAI Painting Brush++ actuates cylindrical grip of brush in radial direction to express reaction force of brushing stroke [Otsuki et al., 2017]. These studies provide intuitive usage of digital 3D drawing with no fixing of device to hand, but they have limitation in DOF of force feedback. An Ungrounded Pen-Shaped Kinesthetic Display provides axial and radial force feedback on pen tip, but it needs to fix the device to index finger with ring fixture [Kamuro et al., 2011].

This paper introduces pen shaped interface with no constraint of device and hand, achieving multi-DOF force feedback, vibrotactile and auditory feedback. We also construct MR interaction system to allow user to draw and interact with virtual objects with proposed pen shaped interface.

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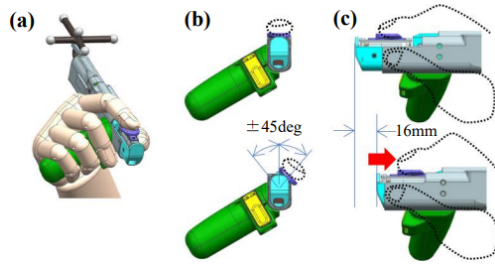


Figure 2: (a) Pen shaped interface held in hand. (b) Rotational force feedback. (c) Linear force feedback.

2 PEN SHAPED INTERFACE

2.1 Hardware configuration

Figure 1 shows the mechanical structure of proposed pen shaped interface. The actuator placed at the tip of pen is used for rotating finger holder to apply rotational force feedback. Another actuator in the rear side of pen drives wire mechanism, which converts rotation to linear motion, to move the pen tip back and forward to apply linear force feedback. Figure 2 (a) shows the pen shaped device held by a user. User will place the index finger on the finger holder and hold side area of pen tip with thumb and middle finger to fix the device firmly. As shown in Figure 2 (b) and (c), force feedback will be applied only on index finger thus user can stabilize the pen with the rest of fingers and palm, without restricting the device to the hand. Linear and rotational force feedback can be applied simultaneously, and the maximum force applied to user is 7.8N in linear force feedback and 4.7N (40.2mNm) in rotational force feedback. Vibrator in the grip, and speaker at the tip of pen are used for vibrotactile and auditory feedback to render texture of virtual object. Pen also contains control switch for user input, and reflective markers for optical tracking described later.

2.2 IMPLEMENTATION

Contact force with the virtual object is calculated based on spring-damper model. The force component vertical to the contact surface will be reflected to linear force feedback, and the force component horizontal to the contact surface will be reflected to rotational force feedback. Texture pattern of virtual object is described as waveform data for vibrator and speaker, and its magnitude and pitch are adjusted based on relative velocity at the contact surface. Physical parameters (spring, damper, friction coefficient, texture pattern) are set independently to each virtual object including drawn lines to express variety of haptic feelings.

3 MR INTERACTION SYSTEM

We developed MR interaction system described in Figure 3 to demonstrate the validity of proposed pen shaped interface. The position of the pen device is tracked by externally placed motion tracking camera system (OptiTrack). As for the display, we chose Spatial Reality Display (ELF-SR1), able to watch volumetric image with bare eyes. User can draw the line and touch the virtual object in the workspace described in Figure 4. Virtual pen tip will be displayed in this area as if the actual pen device is stretched

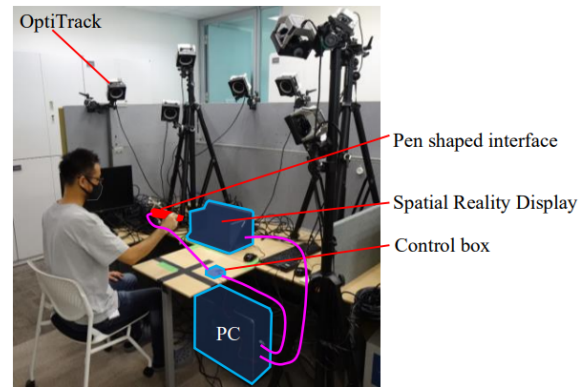


Figure 3: Overview of MR interaction system

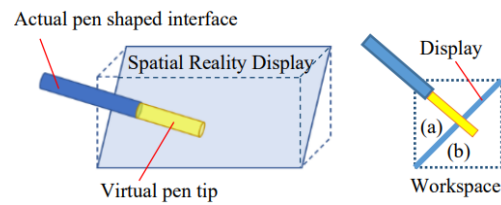


Figure 4: Workspace in MR interaction system. Actual Pen can move in area (a). Virtual pen tip will be displayed in area (a) and (b) as if it is extended from actual pen.

into the virtual environment. We found that the combined use of this visual effect and multimodal feedback from the pen shaped interface will improve the reality of virtual space interaction and provide immersive experience to users.

4 LIMITATION AND FUTURE WORK

The main limitation of our study is that this device can only express horizontal component in terms of tangential force. An additional actuator is required to add vertical tangential force expression. For the future work, we will implement an inside-out position tracking system, battery, and control board inside the pen to create the standalone wireless system. In addition to this improvement, we can use AR glasses or HMD instead of fixed display to remove the limitation of workspace.

REFERENCES

- Hesham Elsayed, Mayra Donaji Barrera, Christian Schaarschmidt, Jan Riemann, Andrii Matvienko, Martin Schmitz, Martin Weigel, and Max Mühlhäuser. 2020. VRSketch-Pen: Unconstrained Haptic Assistance for Sketching in Virtual 3D Environments. In ACM VRST 2020.
- Mai Otsuki, Kenji Sugihara, Azusa Toda, Fumihisa Shibata, and Asako Kimura. 2017. A brush device with visual and haptic feedback for virtual painting of 3D virtual objects. Virtual Reality, Springer-Verlag London 2017. DOI:10.1007/s10055-017-0317-0
- Sho Kamuro, Kouta Minamizawa, and Susumu Tachi. 2011. An Ungrounded Pen-Shaped Kinesthetic Display: Device Construction and Applications. In IEEE World Haptics Conference 2011, 557-562.